

Variational inference for the per-individual mixture SER, and reading off the model verdict (mixture vs. additive/dominant/recessive)

derivation note for CHIMERA / SuSiE-Mix (SuSiE-PIM)

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1 Model: one variant, per-individual mixture

Fix a candidate variant with codings x_i^A, x_i^D, x_i^R (centred), $m \in \{A, D, R\}$, residual variance σ^2 supplied by SuSiE. Latents: a per-individual coding type z_i and effects $\boldsymbol{\beta} = (\beta_A, \beta_D, \beta_R)$, mixing weights $\boldsymbol{\pi}$.

$$\boldsymbol{\pi} \sim \text{Dirichlet}(\mathbf{a}_0), \quad z_i \mid \boldsymbol{\pi} \sim \text{Cat}(\boldsymbol{\pi}), \quad \beta_m \sim \mathcal{N}(0, \sigma_{0m}^2), \quad y_i \mid z_i=m, \boldsymbol{\beta} \sim \mathcal{N}(x_i^m \beta_m, \sigma^2). \quad (1)$$

Write $r_{im} = q(z_i=m)$. We use the mean-field family $q(\mathbf{z}, \boldsymbol{\beta}, \boldsymbol{\pi}) = (\prod_i q(z_i)) (\prod_m q(\beta_m)) q(\boldsymbol{\pi})$ and maximise the ELBO by coordinate ascent (CAVI). Each block update is the exponential of the expected complete-data log-joint with respect to the other factors.

2 CAVI updates

The complete-data log-joint is, up to constants,

$$\log p = \log \text{Dirichlet}(\boldsymbol{\pi}; \mathbf{a}_0) + \sum_m \left(-\frac{\beta_m^2}{2\sigma_{0m}^2} \right) + \sum_i \sum_m \mathbf{1}\{z_i=m\} \left[\log \pi_m - \frac{(y_i - x_i^m \beta_m)^2}{2\sigma^2} \right]. \quad (2)$$

(a) Effects $q(\beta_m) = \mathcal{N}(\mu_m, s_m^2)$. Collecting β_m -terms in $\mathbb{E}_q[\log p]$ (with $\mathbb{E}_q[\mathbf{1}\{z_i=m\}] = r_{im}$): $-\frac{\beta_m^2}{2\sigma_{0m}^2} - \frac{1}{2\sigma^2} \sum_i r_{im} (-2x_i^m y_i \beta_m + (x_i^m)^2 \beta_m^2)$, a Gaussian in β_m , hence

$$s_m^2 = \left(\frac{1}{\sigma_{0m}^2} + \frac{1}{\sigma^2} \sum_i r_{im} (x_i^m)^2 \right)^{-1}, \quad \mu_m = \frac{s_m^2}{\sigma^2} \sum_i r_{im} x_i^m y_i. \quad (3)$$

These are responsibility-weighted Bayesian regressions, one per coding.

(b) Mixing weights $q(\boldsymbol{\pi}) = \text{Dirichlet}(\boldsymbol{\alpha})$. Collecting $\boldsymbol{\pi}$ -terms gives a Dirichlet with

$$\alpha_m = a_{0m} + \sum_i r_{im}, \quad \mathbb{E}_q[\log \pi_m] = \psi(\alpha_m) - \psi(\sum_{m'} \alpha_{m'}). \quad (4)$$

(c) Responsibilities $q(z_i)$. $\log r_{im} = \mathbb{E}_q[\log \pi_m] + \mathbb{E}_q[\log \mathcal{N}(y_i; x_i^m \beta_m, \sigma^2)] + \text{const.}$ Using $\mathbb{E}_q[\beta_m] = \mu_m$, $\mathbb{E}_q[\beta_m^2] = \mu_m^2 + s_m^2$ and dropping i -constant terms (y_i^2 , $\psi(\sum \alpha)$),

$$r_{im} \propto \exp \left\{ \psi(\alpha_m) - \frac{1}{2\sigma^2} (-2x_i^m y_i \mu_m + (x_i^m)^2 (\mu_m^2 + s_m^2)) \right\}, \quad \sum_m r_{im} = 1. \quad (5)$$

(d) Empirical-Bayes prior variance. Maximising the ELBO over σ_{0m}^2 gives the closed update $\sigma_{0m}^2 = \mathbb{E}_q[\beta_m^2] = \mu_m^2 + s_m^2$.

ELBO. Iterating (a)–(d) monotonically increases

$$\begin{aligned} \mathcal{L} = & \sum_i \sum_m r_{im} \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (y_i^2 - 2x_i^m y_i \mu_m + (x_i^m)^2 (\mu_m^2 + s_m^2)) \right] \\ & + \underbrace{\sum_i \sum_m r_{im} (\psi(\alpha_m) - \psi(\alpha_0)) - \sum_i \sum_m r_{im} \log r_{im} - \sum_m \text{KL}(\mathcal{N}(\mu_m, s_m^2) \parallel \mathcal{N}(0, \sigma_{0m}^2)) - \text{KL}(\text{Dirichlet}(\boldsymbol{\alpha}) \parallel \text{Dirichlet}(\boldsymbol{\alpha}_0))}_{z \text{ prior cross-term} + \text{entropy}} \end{aligned} \quad (6)$$

with $\alpha_0 = \sum_m \alpha_m$, the Gaussian KL $\frac{1}{2}[\log(\sigma_{0m}^2/s_m^2) + (s_m^2 + \mu_m^2)/\sigma_{0m}^2 - 1]$, and the Dirichlet KL $\log \frac{B(\mathbf{a}_0)}{B(\boldsymbol{\alpha})} + \sum_m (\alpha_m - a_{0m})(\psi(\alpha_m) - \psi(\alpha_0))$. (Monotone increase verified numerically; $\mathbb{E}[\boldsymbol{\pi}] = \boldsymbol{\alpha}/\alpha_0$ and the μ_m recover the data-generating values.)

3 From one variant to the SER, and the model verdict

For each variant j we score *four* sub-models and combine them.

- Pure $m \in \{A, D, R\}$: closed-form single-coding log marginal likelihood ℓ_{jm}^{pure} (the standard SER term).
- Mixture: the converged ELBO $\mathcal{L}_j$ of (6), a lower bound on the mixture log marginal likelihood (so the verdict is conservative: it never over-declares a mixture).

With sub-model prior weights $\mathbf{w} = (w_A, w_D, w_R, w_{\text{mix}})$ and $\boldsymbol{\ell}_j = (\ell_{jA}^{\text{pure}}, \ell_{jD}^{\text{pure}}, \ell_{jR}^{\text{pure}}, \mathcal{L}_j)$:

$$\text{variant log-ML} \quad L_j = \log \sum_k w_k e^{\ell_{jk}}, \quad (7)$$

$$\text{model posterior} \quad P(\text{model } k \mid j) = \frac{w_k e^{\ell_{jk}}}{\sum_{k'} w_{k'} e^{\ell_{jk'}}}, \quad (8)$$

$$\text{variant posterior (SER output)} \quad \alpha_j \propto \rho_j e^{L_j - \ell^{\text{null}}}, \quad \sum_j \alpha_j = 1. \quad (9)$$

The **verdict** for the selected variant is $\arg \max_k P(\text{model } k \mid j)$: “mixture” iff $k = \text{mix}$, otherwise the winning pure coding (additive / dominant / recessive); when a mixture is called, report $\mathbb{E}[\boldsymbol{\pi}] = \boldsymbol{\alpha}/\alpha_0$.

IBSS plug-in. The single effect’s posterior-mean contribution (the vector IBSS subtracts) is the model-averaged fit

$$\hat{\mathbf{f}} = \sum_j \alpha_j \left[\sum_m P(\text{pure } m \mid j) \hat{\beta}_{jm}^{\text{pure}} x_{\cdot j}^m + P(\text{mix} \mid j) \sum_m r_{\cdot m}^{(j)} \mu_m^{(j)} x_{\cdot j}^m \right]. \quad (10)$$

Because IBSS only needs $\{\alpha_j\}$ and $\hat{\mathbf{f}}$, this SER drops into the existing loop and yields SuSiE-PIM, while each credible set inherits the verdict and $\mathbb{E}[\boldsymbol{\pi}]$.

Efficiency. The closed-form pure scores are computed for all p variants; the mixture VB is run only for the top K variants by best pure score (others cannot be labelled a mixture). Per-SER cost is then $O(np)$ plus K small CAVI fits, i.e. close to a standard SER.

Identifiability caveat. Additive and dominant codings coincide at $g=1$ and differ only at $g=2$; hence within the non-recessive part of a mixture the additive-vs-dominant split is weakly identified. The *mixture-vs-pure* verdict and the recessive mass are robust; the precise (π_A, π_D) partition should be read with this collinearity in mind.